

Hints & Solutions

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Q1 (4) - Geometry

$$\text{If } \arg\left(\frac{z-2}{z+2}\right) = \frac{\pi}{4}$$

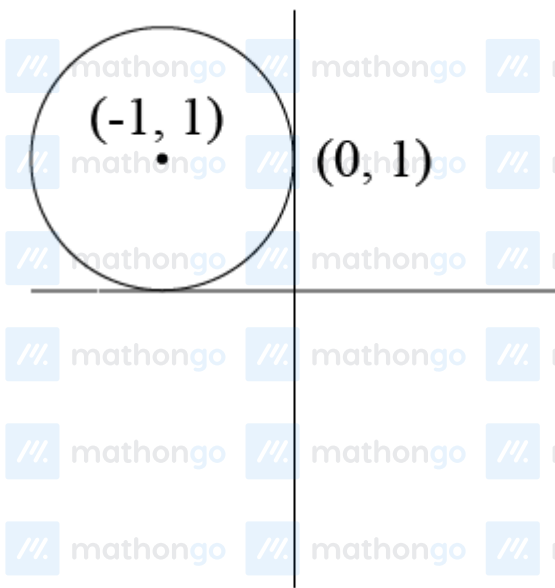
$$\text{i.e. } \tan \frac{\pi}{4} = \frac{4y}{x^2+y^2-4}$$

$$\therefore x^2 + y^2 - 4y - 4 = 0$$

Represents a circle with centre (0,2) and radius $2\sqrt{2}$

Q2 (1) - Geometry

$$|z + 1 - i| \leq 1$$



In the region, least positive argument is $\frac{\pi}{2}$, thus complex number is i .

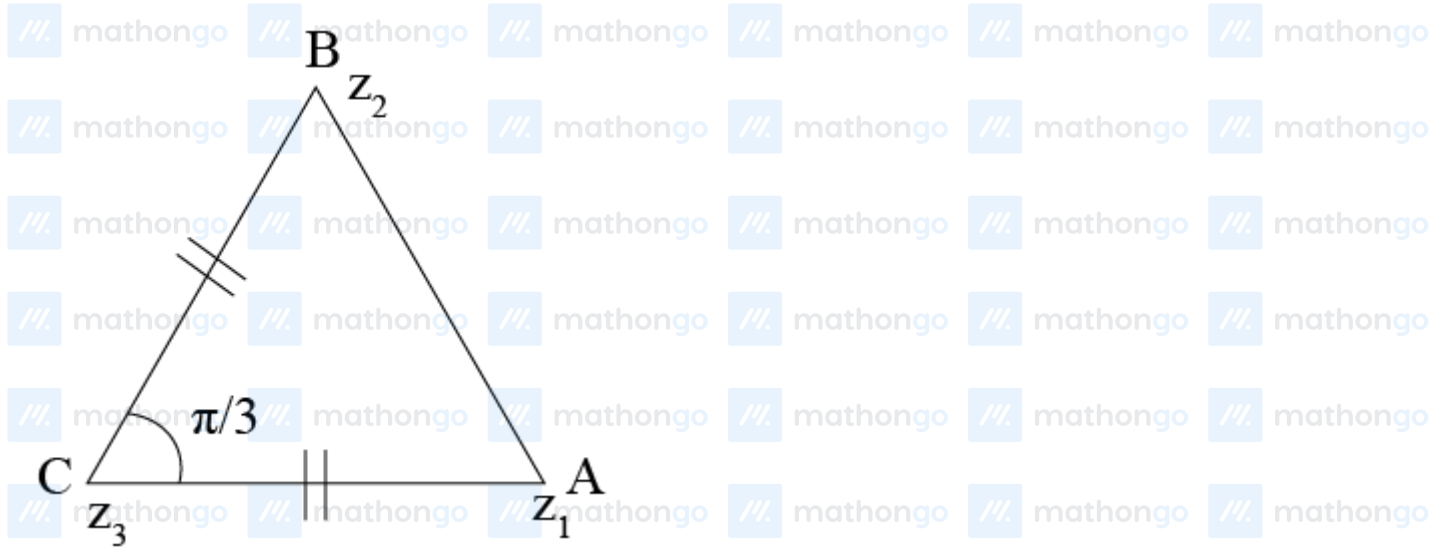
Q3 (3) - Geometry

$$\frac{z_1 - z_3}{z_2 - z_3} = \frac{1 - i\sqrt{3}}{2} = e^{-i\pi/3}$$

$$\text{i.e. } |z_1 - z_3| = |z_2 - z_3| \text{ and angle with vertices } z_3 = \frac{\pi}{3}$$

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So, triangle made by vertices z_1 , z_2 and z_3 is equilateral triangle.

Q4 (4) - Geometry

We have, $|zw| = 1$

$$\therefore \arg(z) - \arg(w) = \frac{\pi}{2}$$

$$\Rightarrow \arg\left(\frac{z}{w}\right) = \frac{\pi}{2}$$

$$\Rightarrow \frac{z}{w} = \left|\frac{z}{w}\right| e^{i\pi/2} = i \frac{|z|}{|w|}$$

$$\Rightarrow z = iw \frac{|z|}{|w|} \dots (i)$$

$$\text{Now } \bar{z}w = -i\bar{w} \frac{|z|}{|w|} \cdot w$$

$$\left[\text{as } \bar{z} = -i\bar{w} \frac{|z|}{|w|}, \text{ by Eq. (i)} \right]$$

$$= -i \frac{|w|^2 |z|}{|w|} = -i|w||z|$$

$$\therefore \bar{z}w = -i$$

Q5 (3) - Geometry

Given, $\bar{z} + i\bar{w} = 0$ and $\arg(zw) = \pi$

$$\Rightarrow \arg(z) + \arg(w) = \pi \dots (i)$$

$$\Rightarrow \bar{z} = -i\bar{w}$$

$$\therefore z = iw \Rightarrow \arg(z) = \frac{\pi}{2} + \arg w$$

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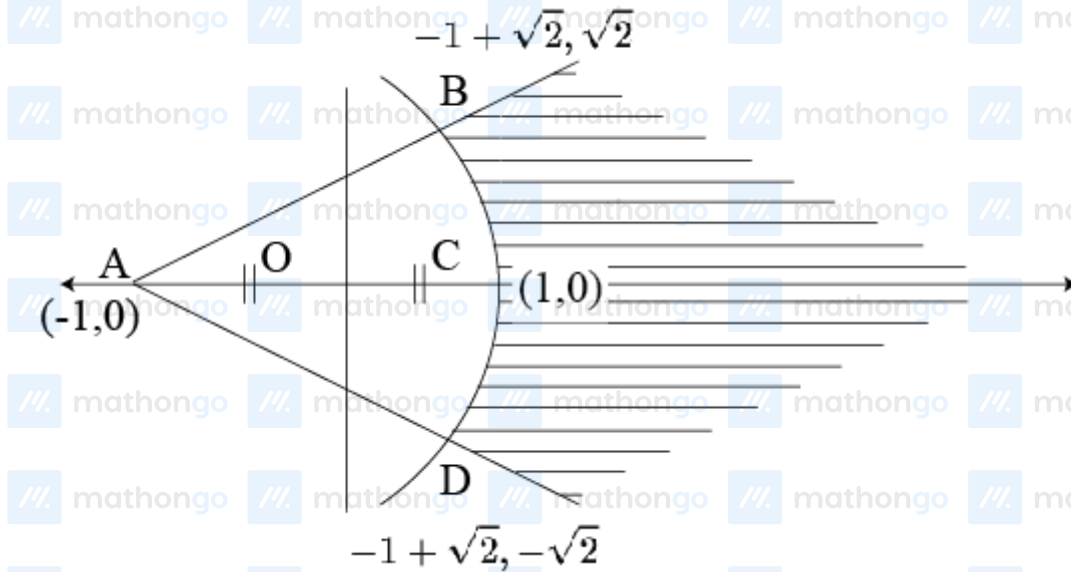
$$\Rightarrow \arg(z) - \arg(w) = \frac{\pi}{2} \dots(ii)$$

$$\Rightarrow 2\arg(z) = \frac{3\pi}{2}$$

$$\therefore \arg(z) = \frac{3\pi}{4}$$

Q6 (1) - Geometry

Region shaded is given by expression of circle



$$\therefore |z + 1| > 2 \text{ and } |\arg(z + 1)| < \frac{\pi}{4}$$

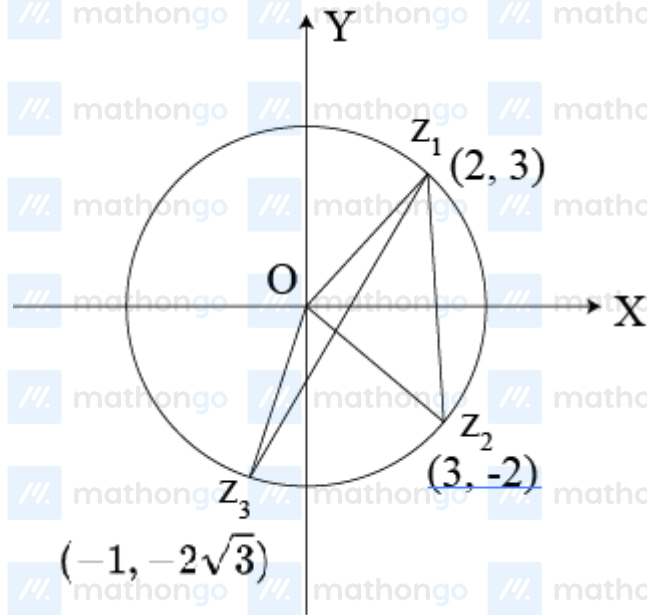
Q7 (3) - Geometry

Here, $|z_1| = |z_2| = |z_3| = \sqrt{13}$

Hence, z_1, z_2, z_3 lies on circle with centre (0,0) and radius $= \sqrt{13}$ as shown.

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$$\text{Now, } \arg\left(\frac{z_2}{z_3}\right) = 2 \arg\left(\frac{z_2 - z_1}{z_3 - z_1}\right)$$

$$\therefore \arg\left(\frac{z_3}{z_2}\right) = 2 \arg\left(\frac{z_3 - z_1}{z_2 - z_1}\right)$$

Q8 (3) - Geometry

Let, $z = e^{i\theta}$ as $|z| = 1$

$$\text{Then, } \left| \frac{z}{z} + \frac{\bar{z}}{z} \right| = 1 \Rightarrow (e^{2i\theta} + e^{-2i\theta}) = 1$$

$$\Rightarrow (2 \cos 2\theta) = 1 \Rightarrow \cos 2\theta = \pm \frac{1}{2}$$

$$\Rightarrow \cos 2\theta = \frac{1}{2} \quad \text{or} \quad \cos 2\theta = -\frac{1}{2}$$

$$\Rightarrow 2\theta = 2n\pi \pm \frac{\pi}{3}$$

$$\Rightarrow \theta = n\pi \pm \frac{\pi}{6}$$

$$\text{and } 2\theta = 2n\pi \pm \frac{2\pi}{3}$$

$$\Rightarrow \theta = n\pi \pm \frac{\pi}{3}$$

For different values of n ,

$$\theta = \frac{\pi}{6}, \frac{7\pi}{6}, \frac{5\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{3}, \frac{4\pi}{3}, \frac{2\pi}{3}, \frac{5\pi}{3}$$

i.e. eight values of z are possible.

Q9 (4) - Nth root of unity

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$$x^n + a_1 x^{n-1} + a_2 x^{n-2} + \dots + a_{n-1} x + a_n = 0$$

$$\Rightarrow 1 + a_1 x^{-1} + a_2 x^{-2} + \dots + a_{n-1} x^{-n+1} + a_n x^{-n} = 0$$

Since, $\cos \theta + i \sin \theta$ is the root

Therefore, $1 + a_1(\cos \theta + i \sin \theta)^{-1} + a_2(\cos \theta + i \sin \theta)^{-2} + \dots + a_{n-1}(\cos \theta + i \sin \theta)^{-n+1} + a_n(\cos \theta + i \sin \theta)^{-n} = 0$

$$\Rightarrow 1 + a_1(\cos \theta - i \sin \theta) + a_2(\cos 2\theta - i \sin 2\theta) + \dots + a_{n-1}(\cos(n-1)\theta - i \sin(n-1)\theta) + a_n(\cos n\theta - i \sin n\theta) = 0$$

On comparing real parts, we get

$$a_1(\cos \theta) + a_2(\cos 2\theta) + \dots + a_{n-1}(\cos(n-1)\theta) + a_n(\cos n\theta) = -1$$

Therefore, $\sum_{i=1}^n a_i \cos i\theta = -1$

Q10 (4) - Nth root of unity

Given, $z + \frac{1}{z} = 2 \cos 13^\circ \Rightarrow z = \cos 13^\circ + i \sin 13^\circ$

$$\Rightarrow z^2 + \frac{1}{z^2} = 2 \cos 26^\circ = 2 \cos B$$

$$\therefore B = 26^\circ$$

Q11 (2) - Nth root of unity

Given, $\alpha = e^{i\frac{2\pi}{n}}$, i.e. α is n th roots of unity i.e. af equation $x^n - 1 = 0$

Then, $x^n - 1 = (x - 1)(x - \alpha)(x - \alpha^2) \dots (x - \alpha^{n-1})$

$$\Rightarrow \frac{x^n - 1}{x - 1} = (x - \alpha)(x - \alpha^2) \dots (x - \alpha^{n-1})$$

Put $x = 11$, then

$$\frac{11^n - 1}{10} = (11 - \alpha)(11 - \alpha^2) \dots (11 - \alpha^{n-1})$$

Q12 (1) - Nth root of unity

Given, $z + \frac{1}{z} = 2 \cos 3^\circ$

$$\Rightarrow z = \cos 3^\circ + i \sin 3^\circ$$

$$\therefore z^{2000} + \frac{1}{z^{2000}}$$

$$= (\cos 6000^\circ + i \sin 6000^\circ) + (\cos 6000^\circ - i \sin 6000^\circ)$$

$$= 2 \cos 6000^\circ = 2 \cos(33 \times 180^\circ + 60^\circ) = -2 \cos 60^\circ = -1$$

$$\Rightarrow z^{2000} + \frac{1}{z^{2000}} + 1 = 0$$

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Q13 (3) - Nth root of unity

Given that

$$x^{2m} + x^m + 1 \text{ is divisible by } x^2 + x + 1$$

 $x^{2m} + x^m + 1$ can be written as

$$x^{2m} + x^m + 1 = (x^2)^m + (x)^m + 1^m$$

Here when $m = 1$ Clearly $x^{2m} + x^m + 1$ becomes $x^2 + x + 1$. Then it is divisible by $x^2 + x + 1$ so m can be 1So in general m cannot be divided by 3

So here in this multiple choice question the answer is

 m is an odd integer and in general cannot be divided by 3.

Q14 (2) - Nth root of unity

$$(x-1) \left(x + \frac{1}{2} - i\frac{\sqrt{3}}{2} \right) \left(x + \frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \\ = (x-1)(x-\omega)(x-\omega^2) = x^3 - 1$$

Q15 (3) - Nth root of unity

Given, $z^2 + z + 1 = 0$

$$\Rightarrow z = \omega \text{ or } \omega^2$$

$$\therefore \left(z + \frac{1}{z} \right)^2 + \left(z^2 + \frac{1}{z^2} \right)^2 + \left(z^3 + \frac{1}{z^3} \right)^2 + \dots + \left(z^6 + \frac{1}{z^6} \right)^2 \\ = (-1)^2 + (-1)^2 + (2)^2 + (-1)^2 + (-1)^2 + (2)^2 = 12$$

Q16 (1) - Nth root of unity

Given, $z^3 + w^5 = 0$ and $z^2 \bar{w}^4 = 1$

$$\Rightarrow z^3 = -w^5 \text{ and } z^2 = \frac{1}{\bar{w}^4}$$

$$\Rightarrow |z|^3 = |w|^5 \text{ and } |z|^2 = \frac{1}{|w|^4}$$

$$\Rightarrow |z|^6 = |w|^{10} \dots (i)$$

$$\text{and } |z|^6 = \frac{1}{|w|^{12}} \dots (ii)$$

From Eqs. (i) and (ii), we get

$$|w|^{10} = \frac{1}{|w|^{12}} \Rightarrow |w|^{22} = 1$$

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$$\Rightarrow |w| = 1 \text{ and } |z| = 1$$

$$\Rightarrow w\bar{w} = 1 \text{ and } z \cdot \bar{z} = 1 \dots (\text{iii})$$

$$\text{Also, } z^6 = w^{10} \text{ and } z^6 \bar{w}^{12} = 1$$

$$z^6 = \frac{1}{\bar{w}^{12}}$$

$$\therefore w^{10} = \frac{1}{\bar{w}^{12}}$$

$$\Rightarrow w^{10} \cdot \bar{w}^{12} = 1$$

$$\Rightarrow w^{10} \cdot \bar{w}^{10} \cdot \bar{w}^2 = 1 \Rightarrow \bar{w}^2 = 1$$

$$\Rightarrow \bar{w} = 10r - 1 \Rightarrow w = 1, -1$$

$$\text{If } w = 1, \text{ then } z^3 + w^5 = 0 \text{ and } z^2 \bar{w}^4 = 1$$

$$\Rightarrow z^3 + 1 = 0 \text{ and } z^2 = 1 \Rightarrow z = -1$$

$$\text{If } w = -1, \text{ then } z^3 + w^5 = 0 \text{ and } z^2 \bar{w}^4 = 1$$

$$\Rightarrow z^3 - 1 = 0 \text{ and } z^2 = 1$$

$$\Rightarrow z = 1$$

Hence, z and w both are purely real

Q17 (4) - Nth root of unity

Let $z = \cos \theta + i \sin \theta$, then

$$\sum_{m=1}^{15} \operatorname{Im}(z^{2m-1}) = \sin \theta + \sin 3\theta + \dots + \sin 29\theta = \frac{\sin 15\theta \sin 15\theta}{\sin \theta}$$

$$= \frac{\sin 30^\circ \sin 30^\circ}{\sin 2^\circ} = \frac{1}{4 \sin 2^\circ} \quad [\because \theta = 2^\circ]$$

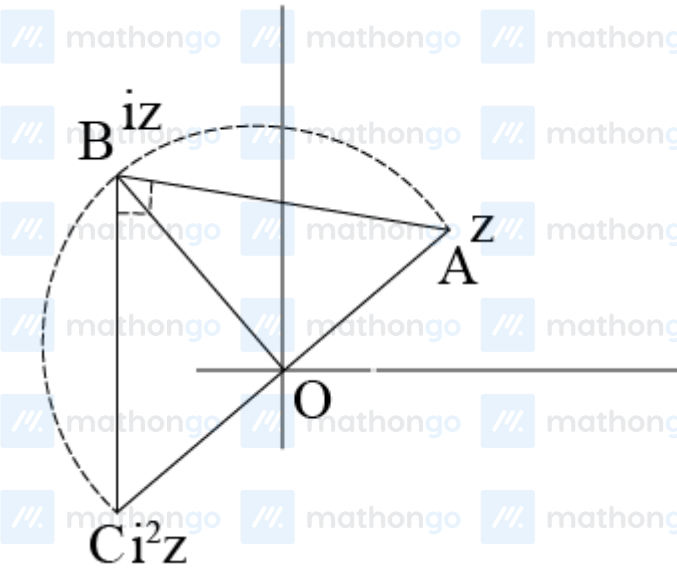
Q18 (4) - Rotation of complex numbers

Triangle formed by z, iz and $i^2 z$ means z is rotated through $\frac{\pi}{2}$ two times i.e. vertices of triangle are along circumference of circle With end point of diameter as two vertices. So, iz is right angled and as in figure $|AB| = |BC|$.

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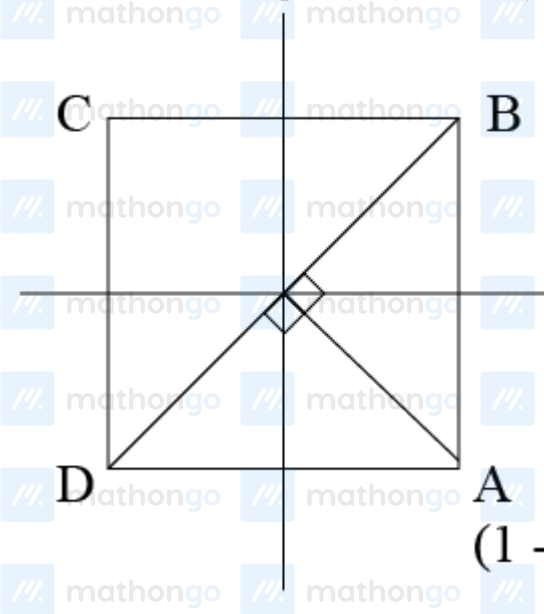
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So, vertices formed right angled isosceles triangle.



Q19 (1) - Rotation of complex numbers

Let $z = 1 - i = \sqrt{2} \left[\cos\left(\frac{-\pi}{4}\right) + i \sin\left(\frac{-\pi}{4}\right) \right] = \sqrt{2}e^{-i\pi/4}$ is vertex at A.



Then, other vertices along other diagonal can be found by rotating it through $\pi/2$ in anti-clockwise and clockwise direction.

i.e.

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$$\sqrt{2}e^{-i\pi/4+\pi/2i} \text{ and } \sqrt{2}e^{-i(\pi/4+\pi/2)}$$

$$\Rightarrow = \sqrt{2}e^{i\pi/4} \text{ and } \sqrt{2}e^{-i3\pi/4}$$

$$\Rightarrow = (1+i) \text{ and } (-1-i)$$

Q20 (1) - Rotation of complex numbers

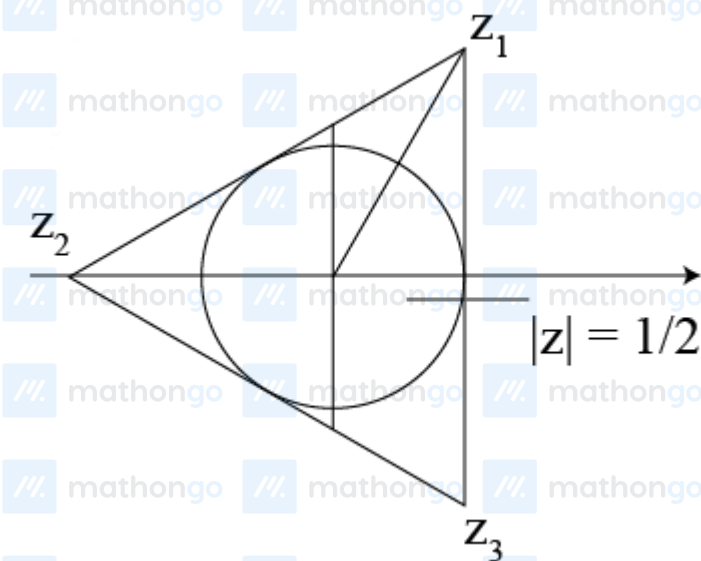
According to the concepts of n th roots of unity, we get all vertices can be determined by rotating one vertex through angle $\frac{2\pi}{n}$ in clockwise or anti-clockwise direction.

If z_1 is one vertex, then z_2 adjacent of z_1 is given by

$$z_2 = z_1 e^{\pm i\frac{2\pi}{n}} = z_1 \left(\cos \frac{2\pi}{n} \pm i \sin \frac{2\pi}{n} \right)$$

Q21 (4) - Rotation of complex numbers

If z_1, z_2, z_3 are vertices of equilateral triangle circumscribing the circle $|z| = \frac{1}{2}$ i.e. z_1 , therefore to obtain z_2 rotated through angle $\frac{2\pi}{3}$ in anti-clockwise direction.



$$\begin{aligned} \text{i.e. } z_2 &= z_1 \cdot e^{i2\pi/3} = \left(\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) e^{i2\pi/3} \\ &= e^{i\pi/3} \cdot e^{i2\pi/3} = e^{i\pi} = \cos \pi + i \sin \pi \\ &= -1 \end{aligned}$$

Q22 (2) - Rotation of complex numbers

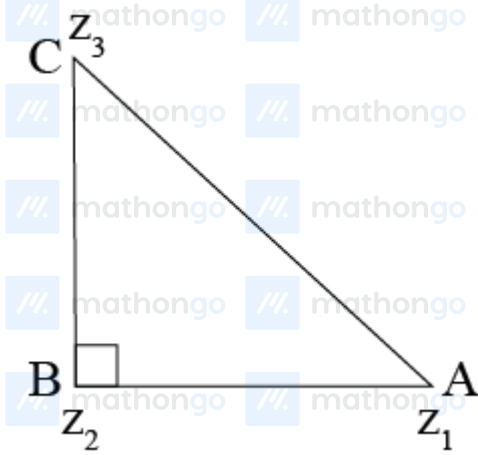
If z and z are vertices of an isosceles triangle and right angled at z_2 i.e. $|z_3 - z_2| = |z_1 - z_2|$

$$\text{i.e. } \frac{z_3 - z_2}{z_1 - z_2} = \frac{|z_3 - z_2|}{|z_1 - z_2|} e^{i\pi/2}$$

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$$\Rightarrow z_2 - z_2 = i(z_1 - z_2)$$

$$\Rightarrow z_3^2 + z_2^2 - 2z_2z_3 = -(z_1^2 + z_2^2 - 2z_1z_2)$$

$$\Rightarrow z_1^2 + 2z_2^2 + z_3^2 = 2z_1z_2 + 2z_2z_3$$

$$\Rightarrow z_1^2 + 2z_2^2 + z_3^2 = 2z_2(z_1 + z_3)$$

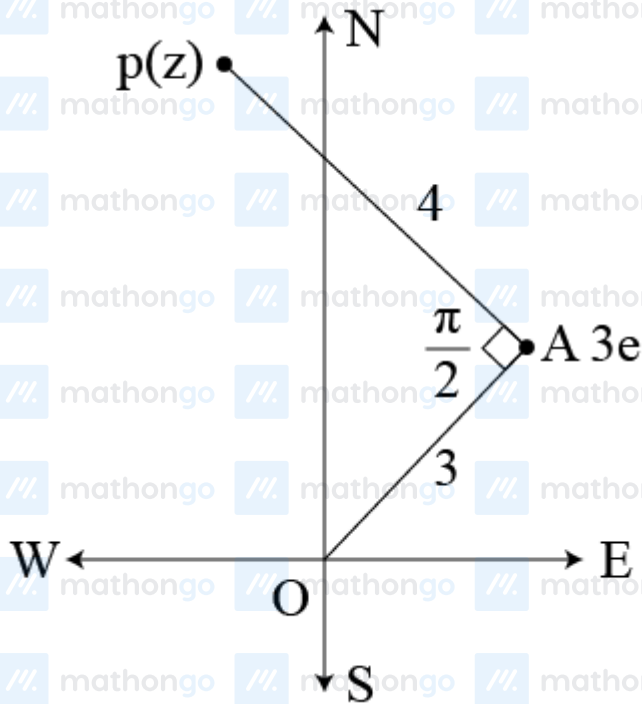
Q23 (4) - Rotation of complex numbers

$$\frac{z-3e^{i\pi/4}}{0-3e^{i\pi/4}} = \frac{4}{3}e^{-i\pi/2}$$

$$= -\frac{4}{3}i$$

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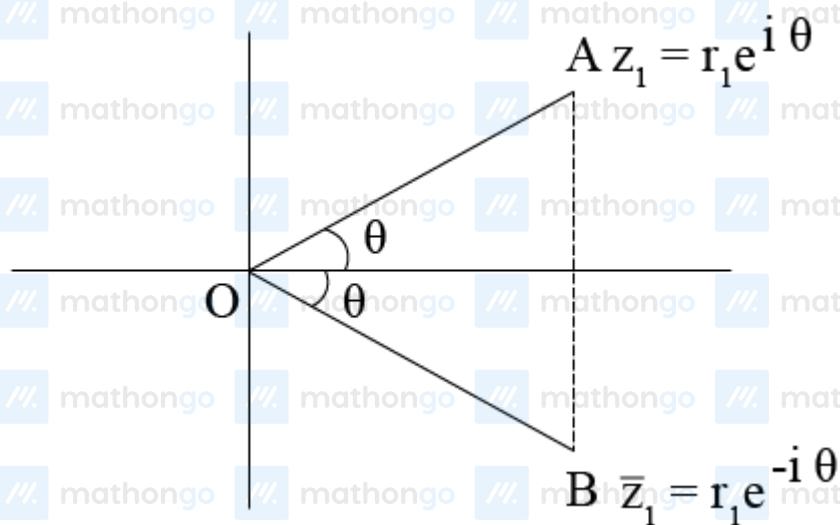


$$\begin{aligned} \therefore z - 3e^{i\pi/4} &= 4e^{i\pi/4} \\ &= 4ie^{i\pi/4} + 3e^{i\pi/4} \\ &= (3 + 4i)e^{i\pi/4} \end{aligned}$$

Q24 (1) - Rotation of complex numbers

If z_1 and $\bar{z}_1$ are adjacent vertices of a regular polygon of n sides with centre at origin and

$$\frac{\text{Im}(z_1)}{\text{Re}(z_1)} = \sqrt{2} - 1 = \tan[\arg(z_1)]$$



$$\text{If } z_1 = r_1 e^{i\theta}, \text{ then } \tan \theta = \sqrt{2} - 1 = \tan(\pi/8)$$

$$\theta = \pi/8$$

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Now rotation of z_1 by 2θ i.e. $\pi/4$ gives adjacent vertex of polygon.

i.e. there will be $\frac{2\pi}{\pi/4} = 8$ sides in polygon. $\bar{A}$

Q25 (2) - Rotation of complex numbers

Straight lines are perpendicular with complex slopes w_1 and w_2 , if $w_1 + w_2 = 0$